Image

2	4	9	1	4	3
2	1	4	4	6	5
1	1	2	9	2	3
7	3	5	1	3	4
2	3	4	8	5	7
4	5	6	7	8	9

Filter

1	2	3
-4	7	4
2	-5	1

2*1+4*2+9*3+2*(-4)+1*7+4*4+1*2+1*(-5)+2*2=51

4*1+9*2+1*3+1*(-4)+4*7+4*4+1*2+2*(-5)+9*2=66

Features

51	66	20	75
31	49	101	12
15	53	-2	53
46	51	61	33

Max Pooling

66	101
53	61

Average Pooling

49	52
41	36

CNN example

In the context of a Convolutional Neural Network (CNN) for image classification, consider an image with a size of 6x6 pixels, a filter of 3x3 and a pooling box of 2x2 dimensions has been provided below.

6	0	2	1	-5	0
-7	-2	0	5	1	-4
0	3	2	-1	0	7
5	1	0	-9	1	6
-4	1	2	0	3	5
-5	2	4	4	1	0

Image

1	0	-5
0	9	0
7	0	2

Kernel / Filter

a) **Calculate** the **output/feature map matrix** after doing one convolution operation using the above kernel on the image. Show all necessary steps to create the map. **[6]**
b) **What** will be the final feature matrix after using average pooling on the convolution matrix calculated in question (b). **[2]**